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A Study on Sensor-based Activity Recognition Having Missing Data

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***Abstract*—Human activity recognition is an important area for various applications. Sensor-based activity recognition deteriorates while partial data are lost. Hence, in this paper, we study activity recognition in the presence of data loss. Earlier, we explored sensor-based activity recognition where we train the data with randomly missed data. It is required to investigate better features for handling missing data. Here, we evaluate activity performance result with missing data environment with various feature combinations for multiple classifiers. Initially, we developed a simulated environment to study the impact of features. Afterward, we evaluated our proposed feature-based method on a benchmark dataset named HASC dataset. The dataset has no missing data. However, to evaluate our approach, we added various levels of missing data randomly and studied the performances. We explored mean, variance, skewness and kurtosis as statistical features based on a time-windowing approach. For classification study, we exploited two classifiers called Naïve Bayes and Random Forest. Our approach and study demonstrated satisfactory recognition results under various feature combinations in different situations of missing data.**

Keywords- Activity recognition, Sensor network, Random Forest, Naïve Bayes

I. INTRODUCTION

Human activity recognition is an important research area in computer vision and sensor-based healthcare monitoring due to various reasons, especially the increase of elderly population and healthcare issues, surveillance, fall detection, etc. [14,15,17]. Activity recognitions are done from video or moving data from various wearable sensors. Different kinds of actions or activities are recognized, e.g., walking, running, sitting, cooking, sports activities, etc. [14]. Recently, wearable sensors become extremely useful in providing accurate and reliable information on people's activities and behaviors, thereby ensuring a safe and sound living environment [2]. Wearable sensors detect normal, abnormal and/or unforeseen situations by monitoring physiological parameters along with other symptoms. Therefore, necessary help can be provided in times of dire need. Machine-to-machine communication has progressed rapidly in healthcare settings [3]. Activities are recognized after evaluating the activity data/video through different machine learning algorithms.

In most of the activity recognition methods, the activity estimation algorithm is automatically generated by supervised machine learning approach. It is evaluated by the training dataset of input sensor data and estimated the correct activity labels. This means that during the data missing environment activity types (activity classes) which do not appear in the training dataset is unable to be estimated when there is no data for specific activity class. This problem is critical when we try to recognizing elderly people's activities, because we cannot estimate the activities for data missing, such as "fall down" or any critical activities.

We studied the missing data issues in feature values [1]. In this paper, we develop a method, which will able to evaluate activity recognition even in missing data environment. We have used two machine learning approaches: Naïve Bayes and Random Forest. Based on a classifier, we build a model and train the dataset by calculating feature values with a particular time stamp data. Then we compute mixtures of several statistical features (i.e., mean, variance, skewness and kurtosis) on the accelerometer data based on specific time-windowing approach. We explored the situations where both test and train data are normal or clean; where only test dataset has missing data; and where missing data are present in both train and test dataset. These combinations have various levels of missing data that are taken randomly. We propose comparatively good feature set for handling missing data during the time of activity recognition.

Initially, we evaluate the performance of our approach, with three different simulation cases without any missing data. Then Missing data is added to the simulated data and we evaluate the performance of our approach. It is for the simple confirmation of our approach. Then we explore a benchmark sensor-based activity recognition dataset, named HASC dataset with and without missing data. We compare the performance in a combination of different features when only testing module have missing data and when train+test both module have missing data.

The paper is organized as follows: after introducing the works in Section I, we present a short background in Section II. In Section III, we present the methodology of our work. We explain the experimental dataset and results in Section IV. A short discussion is presented in Section V. Finally, Section VI concludes the paper with some future work guidelines.

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II. BACKGROUND

Human action or activity recognition is important research areas [14, 15]. These can be accomplished by video-based information or wearable sensor-based data analysis. Recently, the study of assisted living by automatic human activity recognition is highly expected in hospital, elderly care center for improving health and lifestyle. Researchers are struggling to find out regular and unknown activities [16]. Authors in [16] aim to recognize unknown/new activities which do not appear in the training data, combining word vectors constructed from semantic word vectors and Twitter timestamps.

Several researches have been done to predict human activity in many environments and Quality of Service (QoS) of Wireless Sensor Network (WSN). Without optimizing network performance, it will not possible to improve activity recognition accuracy in real time environment. These approaches aim to provide higher reliability, lower end-to-end delay, and higher packet delivery ratio. Authors in [5], reviewed a comprehensive survey of QoS-Aware Routing Protocols in Wireless Body Area Networks. In wireless data transmissions processes, data loss is an important factor that reduces the robustness of wireless sensor networks [8]. However, data loss in wireless sensor networks is a common problem. Data loss has its special patterns due to noise, collision, unreliable link, and unexpected damage, which greatly reduces the accuracy of reconstruction [9]. Earlier, we proposed a method while exploring the recognition performances with missing data in the features [1].

A recent work [4] has described several promising WSN healthcare applications and highlighted the key technical challenges that WSN faces in healthcare. Sensor data loss environment is common issue in wireless sensor network while various sensor data traverse from one source to another. Human activity recognition will not possible properly if we will not able to evaluate all sensors data properly. In the real-world scenario, assisted living facility centers have big challenges to collect all sensor data properly.

It is to be noted that missing data are realistic in wireless network environment. Data can be missed for various reasons, for example, due to network problem, bandwidth limitations, low-energy of IoT devices, server problem, etc. Based on our study, sensor-based activity recognition approaches are accomplished on the available datasets and they did not consider any missing data issue. In this paper, we take this challenge to study the performance of activity recognition while various levels of missing data are present. There are different datasets for sensor-based activity recognition. However, most of these are not available to all. One of the benchmark datasets we used in our work is called HASC dataset [6, 7]. It is the Human Activity Sensing Consortium (HASC) challenge dataset.

III. METHODOLOGY

In sensor-based activity recognition, variety of sensors' data transfer from one source to another with real time sensing capability. Therefore, on time and reliability of data transfer are important issues in a wireless sensor network. Considering the data loss/missing data in a sensor network, we do experiments in random data loss environment and find out activity recognition performance with probable good features set for handling missing data.

Earlier, we studied the recognition performances in missing data environment where the missing parts were in feature values, not in the accelerometer data [1]. Therefore, in this work, we explore the more realistic situation where missing data are actually present in the dataset. As the dataset is created in normal conditions, so, no missing data are available in the HASC dataset or other datasets. Hence, we created various missing data combinations randomly and studied the recognition performances. These are covered in the following sections. Our aim is to handle missing values during activity recognition in multi-sensory networks. There are three levels for handling missing values in such a network; raw data, feature level and classifier. Incomplete data is common in wireless sensor network and may arise due to hardware failures, synchronization, packet collisions, signal strength fading, or environmental interference. For sensor network we consider time series sensor data. A more realistic assumption is that data is missing at random (MAR). The key aspect about MAR is that the values of the missing data can somehow be predicted from some of the other variables being studied. Possible variations are: 'NaN', 'NA', 'None' and others. In our research, we consider random missing data and replace missing value with NA.

In our strategy, our training dataset and test dataset have the presence of missing data. We train with missing data and buildup model with these missing data. Then we test with some random missing data. Figure 1 presents a flow diagram of the method we use in this paper. In this approach, we collect data from sensors. We segment sensor data for testing and training. Data can be extracted from sensors like accelerometer, gyroscope, etc. Afterward, we use windowing approach for feature calculation. For handling missing data in dataset, we need to find out appropriate feature calculation method. We use mean, variance, skewness, and kurtosis for feature calculation. We find that compare to the performances of skewness and kurtosis, mean and variance can distinguish different classes better for this kind of missing data.

By mean value calculation, we summarize the data attributes; while variance can be used in identifying sharp details of time series data. If we consider $X = x_1, x_2, x_3, \ldots, x_n$ are the series of sensor data then the mean value will be, $\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$. Here, n is the number of training instances. The variance of a random variable X is the expected value of the squared deviation from the mean of X: $\mu = E[X]$ and $Var(X) = E[(X-\mu)^2]$.

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Then skewness calculates the asymmetry or skewing rate of the probability distribution of a real-valued random variable about its mean. When we measure the skewness of a symmetrical dataset, it will be equal to 0. So, a normal distribution will have a skewness of 0. On the other hand, kurtosis measures the flatness of a distribution. Kurtosis can define as an average value of the variation of the time series data.

In this approach, missing data are considered in both training and testing modules. However, we study action recognition without having any missing data, as well as, with missing data in the training part. In order to train a machine learning model, we use machine learning algorithm with training data from where it will learn.

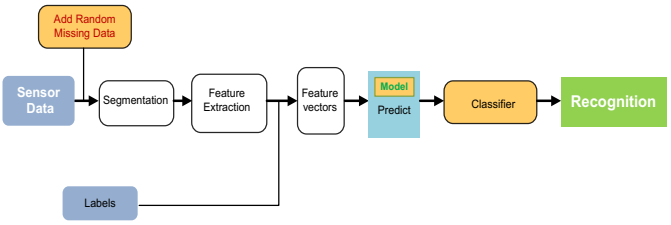


Fig. 1: Flow diagram of our approach having missing data in both training and testing modules.

In this paper, we have used machine learning approaches by using two different classifiers: Naïve Bayes and Random Forest. We first use Naïve Bayes because this algorithm works based on counting and conditional probability [10]. The model can be considered as a probability table. This probability table is updated based on the training information.

Naive Bayes classifier is a supervised learning method. It is also a statistical approach to classify different labels. It is more suitable when the dimensionality of input is high. It performs well in the presence of noise in data.

We also exploit the random forest classifier. Note that random forest is one of the most popular and powerful off-the-shelf classifiers [11]. The random forest classifier is good with regression as well as multiclass classification with better recognition rate [12]. This classifier can produce multiple decision trees, which are developed randomly based on various sampled instances of training data. These trees are trained independently [12]. In the testing process, on the other hand, class labels of decision trees are anticipated by combining the multiple classifiers' predictions [13].

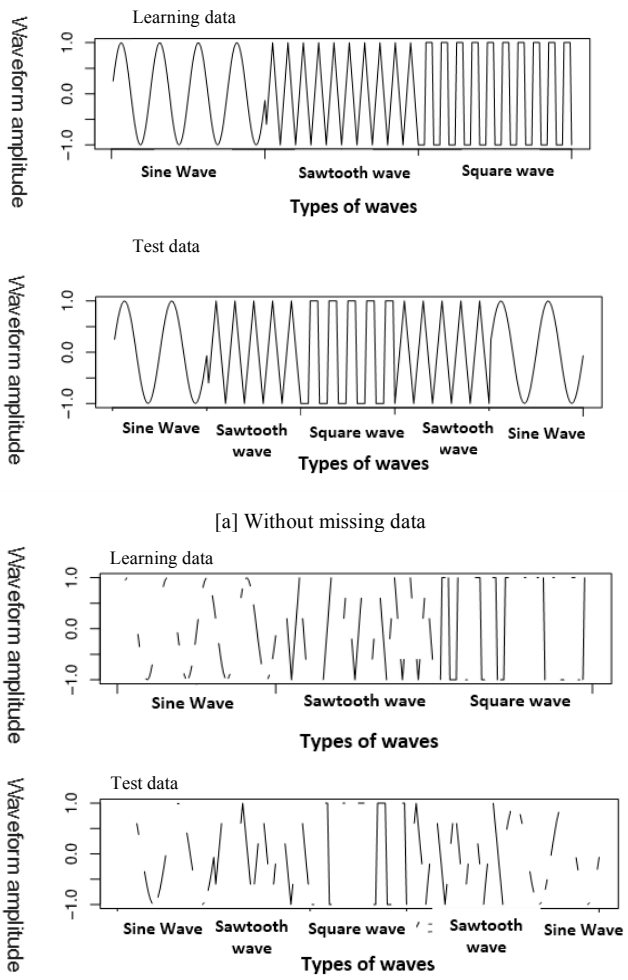
If any missing values remain in the training of decision trees, these missing data could interrupt in proper classification of the decision trees. That is why, the presence of missing data in the training part may produce inaccurate decision tree induction and demonstrate misclassification [12, 13]. Even though a decision tree has some kind of missing values, the combining the multiple classifiers' predictions from many decision trees can decrease the impact of missing values. Therefore, Random Forest can demonstrate better performance even any presence of missing data.

IV. Evaluation

We study the recognition performance under two datasets: One is simulated dataset and a benchmark sensor-based activity dataset called HASC dataset.

(i) Experiment with simulated data:

This is a simple confirmation of classification accuracy. We produce three types of data: sine wave, sawtooth wave, and square wave. Initially, there is no missing value (Fig. 2[a]). Later, we add random missing values in the simulation data (Fig. 2[b]) and evaluate of recognition accuracy with both environments. Figure 3 demonstrates the recognition of these simulated data.



[a] Without missing data

[b] With missing data

Fig. 2: Experiments with simulation data having three kinds of waves: sine wave, sawtooth wave and square wave – [a] Without missing data; and [b] With missing data.

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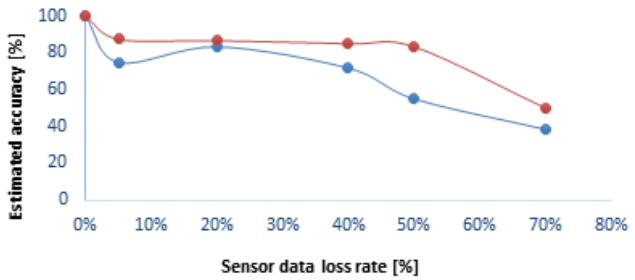


Fig. 3: Recognition performance with simulation data

With simulated data, we observe that when there are no missing data, our recognition accuracy is 100%. Later, we add different percentage random missing data in the testing dataset and in both (training and testing) datasets. We observe that by using our algorithm, we can improve the recognition accuracy when both datasets have missing data. When we add 5% missing data, the recognition accuracy improves from 74.8% to 87.4% by our approach. In this case, the recognition rate is enhanced by 12.6%. However, if we increase the levels of missing data in test or (train + test), the recognition rates reduce gradually.

(ii) Afterward, we explored the HASC dataset in three situations: HASC dataset without any missing data; with missing data in the testing part only; and with missing data in both training and testing modules. The dataset has more than 6700 accelerometer data with 540 subjects have been collected to develop an activity sensor dataset. It has six activities, namely – stay, walk, jog, skip, stair-up, and stair-down. The dataset has two parts: segmented data, and sequence data. The segmented dataset is taken as the training dataset, whereas, the sequence data is considered as the test dataset.

The measurement time of the segmented data is 20 seconds. And number of measurement times is 5 sets. For every activity, data were collected from 10 subjects. Accelerometer data is stored as a simple comma-separated values (CSV) format with time stamp and x, y, z axis-acceleration data. The sequence data have all activities. Each action was done for 10 seconds or more. Total measurement time for this sequence data was 5 minutes from 10 persons. To collect these data, Apple iPod touch 3G is used. Sampling rate was from 10 to 100 Hz. Among the 10 persons, 9 were men and 1 woman. Age range of the subjects was 21 to 32 years. In this dataset, the training set is by accomplished by 10 persons as separate modules, while the testing module has different persons' data. We considered 1-person-left-out cross-validation approach while computing the feature vectors. Training has one after another combinations of different actions and these are labeled properly in a file (*.label). Our this approach can implement with more other activity labels in future in other dataset environment.

We exploited two machine learning approaches (Naïve Bayes and Random Forest) to build a model and train the dataset by calculating feature values with a particular time stamp data, mean and variance of the accelerometer data. Then we injected random missing data in our test dataset, and later added missing data in both train and test dataset. Then, we observed the performance of recognition results in these two environments.

We study mean, variance, skewness, and kurtosis individually to determine possible appropriate feature or feature sets. When we use only the mean and only the variance, we get the recognition accuracy of 39% and 63% by Naïve Bayes classifier respectively, whereas 46% and 61% by Random Forest classifier respectively on the normal clean dataset. Afterward, when we use skewness and kurtosis, we get recognition accuracy of 22% and 20% by Naïve Bayes classifier respectively, whereas 22% and 32% by Random Forest classifier respectively. We find that the impact of skewness and kurtosis are less compare to mean and variance.

Figure 4 is the recognition result of each activity by using Naïve Bayes and Random Forest classifier for mean and variance. We noticed that combination of mean and variance provides relatively better performance. In this result, we observe that all individual activities recognition result is good except 'stDown' activity, which is 0% by using Naïve Bayes and 17% by using Random Forest classifier. However, all other activities recognition rate is more than 93%. Stay and 'stUP' recognition rate is 100% with clean HASC dataset. Average recognition rate is 85% by using Naïve Bayes and 76% by using Random Forest classifier.

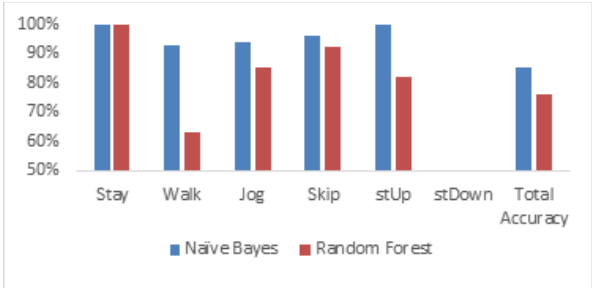
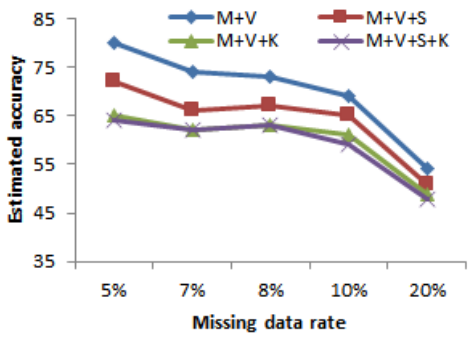


Fig. 4 Accuracy with clean HASC dataset for each activity



[a] Missing data in test module

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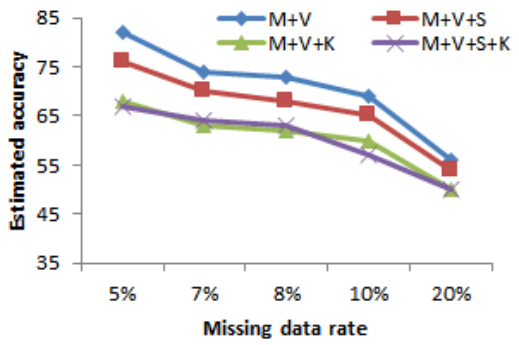
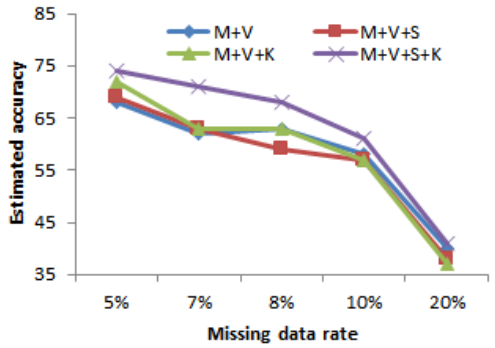
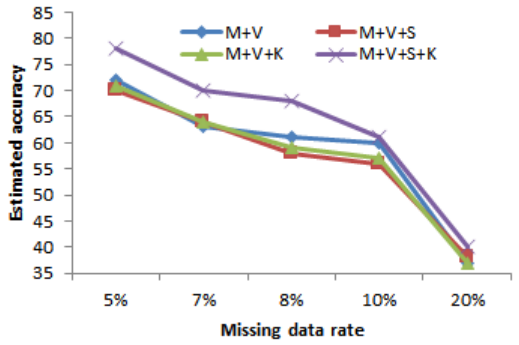


[b] Missing data in test+train module

Fig. 5: Recognition performance for different feature by using Naïve Bayes. Here, 'M','V','S','K means mean, variance, skewness, and kurtosis.



[a] Missing data in test module



[b] Missing data in test+train module

Fig. 6: Recognition performance for different feature by using Random Forest. Here, 'M','V','S','K means mean, variance, skewness, and kurtosis.

For the situation in HASC dataset where missing data are added in the testing part only, we get the following results. Figure 5 and 6 show various recognition results for different combinations of features by using Naïve Bayes classifier and Random Forest classifier.

Figure 5[a] is the result of different features when missing data is only in the test module and Fig. 5[b] shows various recognition results for different combinations of features by using Naïve Bayes classifier when both train and test module has missing data. In the similar fashion, Fig. 6[a] is the result of different features when missing data is only in the test module and Fig. 6[b] shows various recognition results for different combinations of features by using Random Forest classifier when both train and test module has missing data.

Using Naïve Bayes, M-V shows the best results for both cases, whereas, using Random Forest, we get best results with all features. For the HASC dataset where missing data are present in both training and testing modules, we get the results as shown in Figures 5 and 6. In Fig. 5 and 6, we show different results for different features. We find that if we train with missing data and then test with randomly missing data, the performance improves compare to previous one. For the Naïve Bayes classifier, recognition rate becomes 82% with 5% missing data by using mean and variance feature, and 78% with 5% missing data by using Random Forest classifier with all features are combined. Earlier, it was 80% and 74% with 5% missing data in only testing module for Naïve Bayes and Random Forest classifier respectively.

V. Discussions

The simulated data demonstrates some evaluation based on mean and variance. We can notice that Naïve Bayes classifier provides better recognition result than the Random Forest classifier in different percentage of missing data environment with mean and variance feature. It is because that mean and variance are robust for statistical analysis. Initially, we explore one feature at a time to understand the impact of an individual feature on this dataset. Our findings clearly demonstrate that mean and variance provide better cues to extract better feature vector. Later, we explore different combinations of features – with clean data, data having missing part in test module only, and data having missing part in both test and train modules.

Using Naïve Bayes classifier, mean and variance features show a better result. On the other hand, all four features combinedly demonstrate better recognition result using Random Forest classifier. We test in different missing data level. We observe that when we added 5% missing data in testing module then recognition rate is 80% by using Naïve Bayes classifier with mean and variance feature. But in Random Forest classifier better result is 74% by using all features combinedly when only test data has missing data. It gradually decreases based on the amount of missing data in the dataset. In compare to Random Forest, Naïve Bayes result is good.

Then we train the dataset with missing environment and then test with missing data environment. It improves the performance compare to the only missing data in the test dataset. However, this enhanced recognition result is not significant.

During feature analysis time, it has been observed that combination of mean and variance give best result with Naïve Bayes classifier. On the other hand, when we use Random Forest classifier, the combination of mean, variance, skewness, and kurtosis demonstrate the best result. Compare to skewness and kurtosis mean and variance can

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distinguish different classes for this kind of missing data. Finally, we notice that for this dataset and our feature selection, Naïve Bayes provides better recognition result than Random Forest in every situation or feature-combination.

VI. CONCLUSIONS

Activity recognition is an important research topic for various applications, especially in developing sensor-based healthcare monitoring system. Studies are going on for Sensor-based or wearable sensor-based activity recognition. There are challenges to overcome, especially in missing data in the network. There are few works related to missing data in this arena. Hence, in this paper, we explore to propose better features for accelerometer-based data analysis to recognize various actions. In this paper, we study missing data situations. This work can be useful for elderly care center where missing data is crucial for accurate activity recognition. Here, we compare the recognition performance in the presence of different levels of random missing data. We develop three different simulation data with and without any missing data. Then we explore a benchmark dataset which is HASC dataset for our experiment. For the simulation data analysis, both mean and variance are considered. We have analyzed four different types of features, namely mean, variance, skewness, and kurtosis, and their different combinations to recognize various actions of HASC dataset. For both simulation data and real data of HASC dataset, two classifiers are considered for recognition study: Naïve Bayes and Random Forest. Among them, under various experiments of missing data level, mean and variance demonstrate better performance in both classifiers. However, Naïve Bayes provides better recognition result than Random Forest in every case. In future, we need to explore for other features for improved accuracy in missing data environment. Also, we need to incorporate more action classes in sensor-based dataset.

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